AWS Application Load Balancer Project with Auto Scaling

This project demonstrates a highly available and scalable web application architecture on AWS using native AWS services such as Application Load Balancer (ALB), Auto Scaling Group (ASG), Launch Template, AMI, ACM, Route53, and custom VPC networking.

Project Components

• Custom VPC
• Public and Private Subnets
• Internet Gateway
• NAT Gateway
• Bastion Host
• Application Load Balancer (ALB)
• Target Group
• Auto Scaling Group
• Launch Template
• ACM SSL Certificate
• Route53 DNS

Template Instance Setup

A template EC2 instance was created manually to install Apache HTTPD and deploy the application before creating an AMI.

SSL Configuration

A self-signed SSL certificate was configured on the template instance for HTTPS testing. Later, ACM certificates were attached to the ALB for production-grade SSL termination.

AMI and Launch Template

After verifying the application, an AMI was created from the configured instance. A Launch Template was then created using the AMI for Auto Scaling deployments.

Auto Scaling Group

Auto Scaling Group was configured using the Launch Template and deployed across three private subnets for high availability and scalability.

Application Load Balancer

The ALB was deployed in public subnets and connected to backend EC2 instances through a Target Group. HTTPS listener was configured using ACM SSL certificates.

Route53 Configuration

The domain shopping.vpkdevops.online was mapped to the ALB using Route53 Alias records.

Backend Apache HTTPD Installation

```
sudo yum install httpd php-fpm git -y

sudo systemctl restart httpd.service php-fpm.service
sudo systemctl enable httpd.service php-fpm.service

sudo git clone https://github.com/Fujikomalan/aws-elb-site

sudo cp -R aws-elb-site/* /var/www/html/

sudo chown -R apache:apache /var/www/html/*

sudo systemctl restart httpd.service
```

Application Request Flow

```
Client Request
      ↓
Route53 DNS
      ↓
Application Load Balancer (ALB)
      ↓
Target Group
      ↓
Auto Scaling EC2 Instances
      ↓
Apache Web Application
      ↓
Response to Client
```

Security Group Configuration

Security Group	Ports	Source
ALB Security Group	80 / 443	0.0.0.0/0
Backend Security Group	80 / 443	ALB Security Group
Backend Security Group	22	Bastion Security Group
Bastion Security Group	22	My Public IP

AWS Services Used

AWS Service	Purpose
VPC	Custom Networking
EC2	Backend Application Servers
ALB	Load Balancing
Auto Scaling Group	Automatic Scaling
Launch Template	Instance Configuration
AMI	Golden Image
ACM	SSL Certificates
Route53	DNS Management
NAT Gateway	Internet for Private Instances

Conclusion

This project successfully implemented a scalable and secure AWS-native web hosting architecture using Application Load Balancer and Auto Scaling Group. Backend servers were securely deployed inside private subnets while the ALB handled external HTTPS traffic using ACM certificates. The architecture provides high availability, scalability, fault tolerance, and secure application delivery using AWS best practices.